The Greeks of European Options under Black–Scholes

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Abstract

We derive the closed-form expressions for the five classical Greeks $(\Delta, \Gamma, \mathcal{V}, \Theta, \rho)$ of European call and put options under the Black–Scholes model. The derivations rest on a single algebraic identity — the equality of the two normal-density factors that appear when differentiating the price formula — which is established once and reused throughout. We close with the identities that link the Greeks: the Black–Scholes partial differential equation, the Vega–Gamma identity, the conventions for Theta, and the non-uniform convexity of the call price in volatility. The document serves as a reference for the implementation in the project and for the volatility-implied analysis that depends on Vega.

1 Setup

Throughout this document we work under the Black–Scholes model introduced in the companion document `black_scholes.tex`. We fix $S, K > 0$, $r \in \mathbb{R}$, $\sigma > 0$, and $T > t \geq 0$, and let $\tau \stackrel{\text{def}}{=} T - t$. The price of a European call is

$$C^{\text{BS}}(S, K, r, \sigma, \tau) = S\,N(d_1) - Ke^{-r\tau}N(d_2), \tag{1}$$

with

$$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}, \tag{2}$$

where N and φ denote the standard normal CDF and density respectively, and the put price is

$$P^{\text{BS}} = Ke^{-r\tau}N(-d_2) - S\,N(-d_1). \tag{3}$$

A Greek is a partial derivative of the option price with respect to one of its arguments. Each derivative is taken holding the remaining arguments fixed.

Definition 1.1 (The classical Greeks)**.** For a European option with price $V \in \{C^{\text{BS}}, P^{\text{BS}}\}$, the classical Greeks are

$$\Delta = \frac{\partial V}{\partial S}, \qquad \Gamma = \frac{\partial^2 V}{\partial S^2}, \qquad \mathcal{V} = \frac{\partial V}{\partial \sigma}, \qquad \Theta = \frac{\partial V}{\partial t}, \qquad \rho = \frac{\partial V}{\partial r}.$$

The Theta convention is the calendar-time derivative, not the maturity-time derivative; the two differ by a sign because $\partial/\partial t = -\partial/\partial\tau$ when $\tau = T - t$ with T fixed. We use the calendar-time form because it makes the Black–Scholes PDE (Theorem 5.1) read without sign reversals.

Remark 1.2 (Two readings of Greeks). Greeks admit two complementary readings. As *sensitivities*, they describe how the option price moves under marginal changes to the inputs, and form the basis of the dynamic hedging used at every options desk. As *coefficients of a partial differential equation*, they are the building blocks of the no-arbitrage condition the price must satisfy: the Black–Scholes equation is a linear relation among Θ, Γ, Δ and V itself. The algebraic and the dynamical readings are not independent; they are two manifestations of the same underlying replication argument.

2 The Key Identity

The closed forms for the Greeks could in principle be obtained by brute differentiation of (1). Direct computation produces a profusion of terms involving $\partial d_1/\partial x$ and $\partial d_2/\partial x$ for each input x. Almost all of those terms cancel, and the cancellation rests on a single identity which we establish first.

Lemma 2.1 (Density identity)**.** *For all admissible S, K, r, σ, τ,*

$$S\,\varphi(d_1) = Ke^{-r\tau}\,\varphi(d_2). \tag{4}$$

Proof. Compute the difference of the squares of d_1 and d_2, using $d_2 = d_1 - \sigma\sqrt{\tau}$:

$$\begin{aligned} d_1^2 - d_2^2 &= (d_1 - d_2)(d_1 + d_2) = \sigma\sqrt{\tau}\cdot\left(2d_1 - \sigma\sqrt{\tau}\right) \\ &= 2d_1\sigma\sqrt{\tau} - \sigma^2\tau. \end{aligned}$$

Substitute the definition of d_1 into the first term:

$$2d_1\sigma\sqrt{\tau} = 2\log(S/K) + (2r + \sigma^2)\tau.$$

Therefore

$$d_1^2 - d_2^2 = 2\log(S/K) + 2r\tau,$$

which gives

$$\exp\left(\tfrac{1}{2}(d_2^2 - d_1^2)\right) = \frac{Ke^{-r\tau}}{S}.$$

The identity follows by multiplying by S and recognising the density:

$$S\,\varphi(d_1) = S\cdot\frac{1}{\sqrt{2\pi}}e^{-d_1^2/2} = \frac{Ke^{-r\tau}}{\sqrt{2\pi}}e^{-d_2^2/2} = Ke^{-r\tau}\,\varphi(d_2). \qquad \square$$

The identity holds because d_1 and d_2 are not independent; they share the same numerator structure and differ only by $\sigma\sqrt{\tau}$. The symmetric quadratic form $d_1^2 - d_2^2$ collapses to something σ-free, so the imbalance between $\varphi(d_1)$ and $\varphi(d_2)$ is exactly compensated by $Ke^{-r\tau}/S$.

Remark 2.2 (Where the identity comes from). The identity (4) is not a coincidence. It is the algebraic shadow of the change of numéraire used in the derivation of the Black–Scholes formula. Recall from `black_scholes.tex` that $N(d_1) = \mathbb{P}^S(S_T > K)$ under the stock-numéraire measure, while $N(d_2) = \mathbb{Q}(S_T > K)$ under the bond-numéraire (risk-neutral) measure. The two measures differ by the Radon–Nikodym derivative $d\mathbb{P}^S/d\mathbb{Q} = S_T B_t/(B_T S_t)$, and the densities of $\log S_T$ under the two measures shift by exactly the amount that equates $S\varphi(d_1)$ and $Ke^{-r\tau}\varphi(d_2)$ at the strike. The cancellations we will exploit below are the algebraic image of that probabilistic equivalence.

3 The First-Order Greeks

We derive Δ, $\mathcal{V}$, and ρ using Lemma 2.1. The strategy is identical in each case: differentiate (1), isolate the φ-terms, and apply (4) to make them cancel. The remaining terms give the closed form.

3.1 Delta

Proposition 3.1 (Delta)**.**

$$\Delta_C = N(d_1), \qquad \Delta_P = N(d_1) - 1 = -N(-d_1).$$

Proof. Differentiate (1) with respect to S:

$$\Delta_C = N(d_1) + S\varphi(d_1)\frac{\partial d_1}{\partial S} - Ke^{-r\tau}\varphi(d_2)\frac{\partial d_2}{\partial S}.$$

From (2), both $\partial d_1/\partial S$ and $\partial d_2/\partial S$ equal $1/(S\sigma\sqrt{\tau})$ (since $d_2 - d_1$ does not depend on S). Thus

$$\Delta_C = N(d_1) + \frac{1}{S\sigma\sqrt{\tau}}\big[S\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)\big],$$

and the bracketed term vanishes by (4). Therefore $\Delta_C = N(d_1)$. For the put, differentiate (3) or use put-call parity $C - P = S - Ke^{-r\tau}$, which gives $\Delta_C - \Delta_P = 1$. □

Remark 3.2 (Replication interpretation)*.* The Black–Scholes price (1) is structured as $S \cdot N(d_1) - Ke^{-r\tau} \cdot N(d_2)$, which is the value of a portfolio holding $N(d_1)$ shares financed by borrowing $Ke^{-r\tau}N(d_2)$ at the riskless rate. Proposition 3.1 makes this visible: the share quantity $N(d_1)$ is exactly the hedge ratio, the option's sensitivity to the underlying. The fact that $\partial C/\partial S$ lands cleanly on $N(d_1)$ — without trailing φ terms — is not algebraic luck, but the internal consistency of the replication argument that produced the formula in the first place.

3.2 Vega

Proposition 3.3 (Vega)**.**

$$\mathcal{V}_C = \mathcal{V}_P = S\varphi(d_1)\sqrt{\tau}.$$

Proof. Differentiate (1) with respect to σ:

$$\mathcal{V}_C = S\varphi(d_1)\frac{\partial d_1}{\partial \sigma} - Ke^{-r\tau}\varphi(d_2)\frac{\partial d_2}{\partial \sigma}.$$

By (4), the two density factors satisfy $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$, so this collapses to

$$\mathcal{V}_C = S\varphi(d_1)\left(\frac{\partial d_1}{\partial \sigma} - \frac{\partial d_2}{\partial \sigma}\right).$$

From $d_1 - d_2 = \sigma\sqrt{\tau}$, $\partial d_1/\partial\sigma - \partial d_2/\partial\sigma = \sqrt{\tau}$, giving $\mathcal{V}_C = S\varphi(d_1)\sqrt{\tau}$.

For the put, differentiate (3) or apply put-call parity: $\partial(C - P)/\partial\sigma = 0$, since the parity right-hand side does not depend on σ. Hence $\mathcal{V}_P = \mathcal{V}_C$. □

Remark 3.4 (Vega is positive, and identical for calls and puts)*.* Vega is strictly positive because all three factors are. This is the monotonicity that underlies the well-posedness of the implied-volatility inversion (see `implied_volatility.tex`, Lemma 2.1). The equality $\mathcal{V}_C = \mathcal{V}_P$ is not a coincidence either: parity differentiated in σ has zero on the right. The same argument gives $\Gamma_C = \Gamma_P$ and $\Theta_C - \Theta_P = -rKe^{-r\tau}$; only Delta and Rho carry genuine option-type information.

3.3 Rho

Proposition 3.5 (Rho)**.**

$$\rho_C = K\tau e^{-r\tau}N(d_2), \qquad \rho_P = -K\tau e^{-r\tau}N(-d_2).$$

Proof. Differentiate (1) with respect to r:

$$\rho_C = S\varphi(d_1)\frac{\partial d_1}{\partial r} + K\tau e^{-r\tau}N(d_2) - Ke^{-r\tau}\varphi(d_2)\frac{\partial d_2}{\partial r}.$$

Both $\partial d_1/\partial r$ and $\partial d_2/\partial r$ equal $\sqrt{\tau}/\sigma$ (a constant independent of d_1 versus d_2 shift), so $\left[S\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)\right]\cdot\sqrt{\tau}/\sigma$ vanishes again by (4). Only the $K\tau e^{-r\tau}N(d_2)$ term survives. The put result follows from parity. □

4 The Second-Order and Time Greeks

The remaining Greeks — Γ and Θ — arise from second derivatives or from time. Γ gets a separate treatment because it does not require (4); it falls out of differentiating $\Delta = N(d_1)$ directly. Θ involves the most cancellation and is the longest derivation.

4.1 Gamma

Proposition 4.1 (Gamma)**.**

$$\Gamma_C = \Gamma_P = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}}.$$

Proof. By Proposition 3.1, $\Delta_C = N(d_1)$. Differentiating in S:

$$\Gamma_C = \varphi(d_1)\frac{\partial d_1}{\partial S} = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}}.$$

The put case follows from $\Delta_C - \Delta_P = 1$ (constant in S). □

4.2 Theta

Proposition 4.2 (Theta)**.**

$$\Theta_C = -\frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2),$$
$$\Theta_P = -\frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} + rKe^{-r\tau}N(-d_2).$$

Proof. Recall $\tau = T - t$, so $\partial/\partial t = -\partial/\partial\tau$. Differentiate (1) with respect to τ:

$$\frac{\partial C^{\mathrm{BS}}}{\partial\tau} = S\varphi(d_1)\frac{\partial d_1}{\partial\tau} + Kre^{-r\tau}N(d_2) - Ke^{-r\tau}\varphi(d_2)\frac{\partial d_2}{\partial\tau}.$$

The second term is straightforward. For the first and third,

$$\frac{\partial d_1}{\partial\tau} = \frac{r + \frac{1}{2}\sigma^2}{\sigma\sqrt{\tau}} - \frac{d_1}{2\tau}, \qquad \frac{\partial d_2}{\partial\tau} = \frac{r - \frac{1}{2}\sigma^2}{\sigma\sqrt{\tau}} - \frac{d_2}{2\tau}.$$

Group the φ-terms. By Lemma 2.1, $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$, so we factor out a single $S\varphi(d_1)$ from both:

$$S\varphi(d_1)\left[\frac{\partial d_1}{\partial\tau} - \frac{\partial d_2}{\partial\tau}\right] = S\varphi(d_1)\left[\frac{\sigma}{\sqrt{\tau}} + \frac{d_2 - d_1}{2\tau}\right].$$

Since $d_1 - d_2 = \sigma\sqrt{\tau}$, the bracketed expression becomes

$$\frac{\sigma}{\sqrt{\tau}} - \frac{\sigma}{2\sqrt{\tau}} = \frac{\sigma}{2\sqrt{\tau}}.$$

Therefore

$$\frac{\partial C^{\mathrm{BS}}}{\partial \tau} = \frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} + rKe^{-r\tau}N(d_2),$$

and the calendar-time derivative is the negative,

$$\Theta_C = \frac{\partial C^{\mathrm{BS}}}{\partial t} = -\frac{\partial C^{\mathrm{BS}}}{\partial \tau} = -\frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2).$$

The put expression follows from parity differentiated in t: $\Theta_C - \Theta_P = \partial(C-P)/\partial t = \partial(S - Ke^{-r\tau})/\partial t = -rKe^{-r\tau}$. □

Remark 4.3 (Sign of Theta). For a non-dividend-paying call, the leading φ-term is always negative, and the discount term is also negative whenever $N(d_2) > 0$. Hence $\Theta_C < 0$: the call loses value as time passes, all else equal. For the put, the sign of Θ_P is ambiguous because the two terms have opposite signs; deep ITM puts near expiry can have positive Theta when the discount benefit outweighs the time-value decay.

5 Identities and Structural Properties

The closed forms hide several relations among the Greeks. We collect them here because they appear repeatedly in pricing, hedging, and validation.

5.1 The Black–Scholes PDE in Greeks

Theorem 5.1 (Black–Scholes PDE)**.** *The price of a European call (or put) under the Black–Scholes model satisfies*

$$\Theta + \tfrac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta - rV = 0, \tag{5}$$

identically in (S, K, r, σ, τ)*, with* $V \in \{C^{\mathrm{BS}}, P^{\mathrm{BS}}\}$.

Proof. Substituting the closed forms from Propositions 3.1, 4.1, and 4.2 into (5) for the call:

$$\begin{aligned}
&-\frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2) \\
&\quad + \tfrac{1}{2}\sigma^2 S^2 \cdot \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}} + rS\,N(d_1) - r\big[SN(d_1) - Ke^{-r\tau}N(d_2)\big] \\
&= -\frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} + \frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} + rSN(d_1) - rSN(d_1) - rKe^{-r\tau}N(d_2) + rKe^{-r\tau}N(d_2) \\
&= 0.
\end{aligned}$$

The put case follows by parity, or by the same substitution with the put closed forms. □

Remark 5.2 (Why this identity matters). Theorem 5.1 couples four of the five classical Greeks ($\mathcal{V}$ and ρ do not appear because the model treats σ and r as constants). The PDE is therefore a stringent joint test of any closed-form or numerical implementation: a sign error or missing factor in a single Greek breaks the residual at machine precision. The numerical validation suite in `validate_greeks.py` exploits this directly to triangulate correctness.

5.2 The Vega–Gamma identity

Proposition 5.3 (Vega–Gamma identity)**.**

$$\mathcal{V} = S^2 \, \sigma \, \tau \, \Gamma. \tag{6}$$

Proof. Substituting the closed forms, $S^2\sigma\tau \cdot \varphi(d_1)/(S\sigma\sqrt{\tau}) = S\varphi(d_1)\sqrt{\tau} = \mathcal{V}$. □

Remark 5.4 (The dual nature of Vega and Gamma). The Vega–Gamma identity says that, for a vanilla European option, being long Vega and being long Gamma are the same exposure up to a deterministic rescaling. This collapses two notions of risk that the trader's intuition might keep separate. The collapse is specific to Black–Scholes: in stochastic volatility models (Heston, SABR), σ becomes a state variable in its own right, the Vega-equivalent splits into multiple sensitivities to the volatility process, and (6) no longer holds. The Black–Scholes collapse is one more way in which the model is a degenerate special case of the broader theory.

5.3 Curvature in volatility

The first derivative of C^{BS} in σ (Vega) is strictly positive. The second derivative (sometimes called *Volga*) is not.

Proposition 5.5 (Volga)**.**

$$\frac{\partial^2 C^{\mathrm{BS}}}{\partial \sigma^2} = \frac{S\varphi(d_1)\sqrt{\tau}}{\sigma} \, d_1 \, d_2. \tag{7}$$

Proof. Differentiate $\mathcal{V} = S\varphi(d_1)\sqrt{\tau}$ with respect to σ:

$$\frac{\partial \mathcal{V}}{\partial \sigma} = S\sqrt{\tau} \cdot \varphi'(d_1) \cdot \frac{\partial d_1}{\partial \sigma}.$$

Using $\varphi'(x) = -x\varphi(x)$ and $\partial d_1/\partial\sigma = -d_2/\sigma$ (a calculation that isolates the explicit σ-dependence and uses $d_2 = d_1 - \sigma\sqrt{\tau}$):

$$\frac{\partial \mathcal{V}}{\partial \sigma} = S\sqrt{\tau} \cdot (-d_1)\varphi(d_1) \cdot \frac{-d_2}{\sigma} = \frac{S\varphi(d_1)\sqrt{\tau}}{\sigma} \, d_1 d_2.$$ □

The sign of Volga therefore tracks the sign of $d_1 d_2$. For options moderately near at-the-money with small-to-moderate σ, d_1 and d_2 typically straddle zero and the call is locally *concave* in σ; for options far from ATM or for large σ, they share signs and the call is convex. The change of curvature occurs along the locus $d_1 d_2 = 0$, which is the central object of the smile-implied-volatility approximation literature.

Remark 5.6 (Implication for Newton's method on implied volatility). Volga's sign change is the technical reason why monotonicity of C^{BS} in σ is not by itself sufficient for global convergence of Newton's method when computing implied volatility, as discussed in `implied_volatility.tex` (Section 3.2). When the initial guess and the true root lie on opposite sides of the locus $d_1 d_2 = 0$, the first Newton step can overshoot. This is why the implementation uses a region-aware initial guess (Brenner–Subrahmanyam near ATM, neutral default elsewhere) rather than a constant.

6 Summary

Greek	Call	Put
Δ	$N(d_1)$	$N(d_1) - 1$
Γ	$\dfrac{\varphi(d_1)}{S\sigma\sqrt{\tau}}$	
$\mathcal{V}$	$S\varphi(d_1)\sqrt{\tau}$	
Θ	$-\dfrac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2)$	$-\dfrac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} + rKe^{-r\tau}N(-d_2)$
ρ	$K\tau e^{-r\tau}N(d_2)$	$-K\tau e^{-r\tau}N(-d_2)$

The Greeks satisfy three structural identities: the Black–Scholes PDE $\Theta + \frac{1}{2}\sigma^2 S^2\Gamma + rS\Delta - rV = 0$, the Vega–Gamma identity $\mathcal{V} = S^2\sigma\tau\Gamma$, and the non-trivial sign of Volga $\partial\mathcal{V}/\partial\sigma$ which governs the convexity of the price in volatility. Each identity is exploited downstream: the PDE in the validation of the Greeks implementation, the Vega–Gamma identity as an algebraic sanity check, and Volga in the analysis of Newton's method for implied volatility.